

volatility, not on changes to the implied volatility (as the options are held until expiry). By contrast, in the vega sector, only trading strategies involving changes in the implied volatility are sensible (since the option price depends largely on implied volatility) and practicable (since holding until expiry takes too much time).

In order to further minimize the impact from moves in the underlying and thus the need for delta hedging, one should execute option trades in the vega sector by using straddles. By having largely excluded realized volatility as a driving factor, one obtains a pure exposure to implied volatility and can now consider and trade the volatility surface *in abstraction* from the underlying swap rates, by statistical analysis and trading tools.

Option strategy type ③ is therefore constructed using straddles in the vega sector, mainly exposed to changes in implied volatility, and unwound well before expiry.

### Different Exposure of Different Types of Option Trades

This abstract classification of option trades into three different types is important in order to ensure clarity about the exposure and a correct expression of the exposure desired. We stress this point because some analysts tend to confuse the driving factors in option trades and therefore construct option trades whose exposures do not match the ones outlined in their reasoning. As an illustration, we show how similar three “2Y versus 10Y” option trades provide very different exposures:

- ① Buy 6M2Y receivers,<sup>8</sup> sell 6M10Y receivers, and do not delta hedge. Then you have a 2Y-10Y yield curve steepening position, *conditional on a rally*, with no exposure to the continuous realized volatility<sup>9</sup> and no exposure to changes in implied volatility after entering the trade.
- ② Buy 6M2Y receivers, sell 6M10Y receivers, and delta hedge until expiry. Then you have a realized 2Y volatility minus a realized 10Y volatility position, with no exposure to the curve steepness at expiry<sup>10</sup> and no exposure to changes in implied volatility after entering the trade.

<sup>8</sup>This notation refers to a receiver swaption on 2Y swaps with 6M time to expiry.

<sup>9</sup>Except from the part of continuous realized volatility that is reflected in the position of the underlying at the end (i.e. the “1-period-realized volatility”). However, as in the example of the S&P500 index above, it is possible that a high (continuous) volatility results in no change of the underlying between entry and expiry of the option.

<sup>10</sup>Except from the part of overall curve moves that is reflected in continuous realized volatility. A large move in one direction will be accompanied by some (continuous) volatility.

③ Buy 5Y2Y straddles, sell 5Y10Y straddles, and unwind after one month. Then, you have an implied 5Y2Y volatility minus an implied 5Y10Y volatility position (with very little exposure to the curve steepness and very little exposure to realized volatility).

Of course, it is possible to switch categories. For example, starting with ②, after three months of delta hedging, one could decide to stop delta hedging (i.e. to shift to ①) or to unwind the remaining 3M options (i.e. to shift to ③). However, within each category of trades, both the exposure and the corresponding expression/execution are defined and must not be confused with another category.

In the following, we shall discuss these three types of option trades separately. Given their different exposure and construction, they require different analysis tools, which we shall also develop below.

### OPTION TRADE TYPE ①: SINGLE UNDERLYING

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Buying or selling any combination of options on the same underlying and not delta hedging gives an exposure to the payoff profile of that combination of options. Since the payoff of an option depends on the underlying *if it is above or below the strike*, this results in a *conditional* exposure to the yield of the underlying at expiry. For example, buying a 6M2Y receiver with a strike of 1% and not delta hedging it until expiry means entering into a long 2Y swap position *conditional* on the 2Y swap rate in six months' time being below 1%.

In the absence of delta hedging, there are only two driving forces of the P&L of such strategies: the option premium received or paid at entry and the loss or gain from exercising the option at expiry (if it ends ITM).<sup>11</sup> Depending on the payoff profile constructed and the implied volatility at entry, one receives or pays a premium. If one has received a premium at entry (e.g. by selling receiver swaptions), one has to pay at expiry in the event the option ends ITM.

Thus, in order to assess value in option trades of type ①, one needs to compare the two driving forces of their profits and losses: premium at entry and potential payoff at expiry. The simplest way to do so is to calculate the breakeven value of the underlying at expiry (i.e. the value of the underlying at expiry at which the payoff of the option is equal to its initial premium). For example, if we sold an ATMF receiver swaption, we could calculate the

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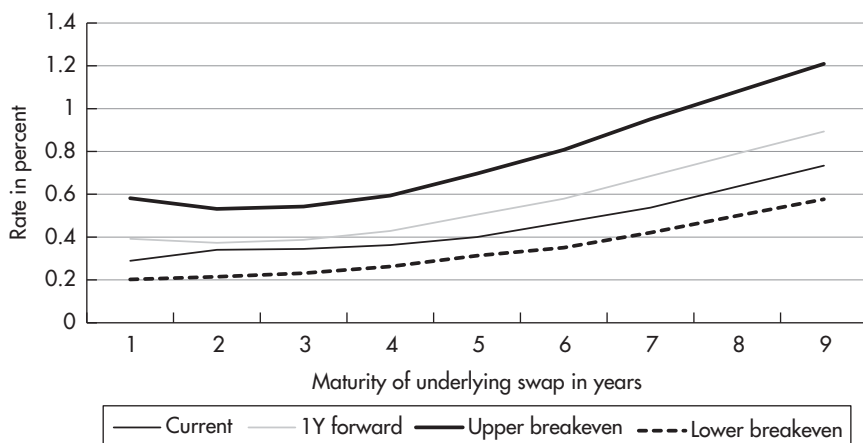
<sup>11</sup>We restrict this analysis to European options. However, similar analysis tools can be used for American options as well.

decline in the underlying swap rate which would cause a loss at expiry that was equal to the premium we had received up-front. If the yield ended above that breakeven level, our overall P&L from that strategy would be positive; if it finished below the breakeven level, it would be negative.

By assessing the breakeven levels of these option strategies, an analyst can see:

- whether expressing a macro view on the yield curve with options (rather than with the underlying) is attractive at all;
- and, if it is, which option strategy (e.g. short receivers or short straddles?) is better;
- and then, once that is decided, which underlying and expiry provide the best expected return.

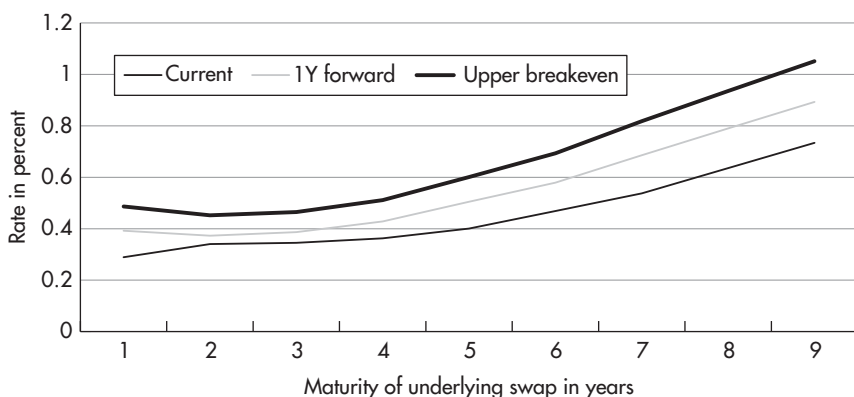
In order to support that analysis, the basic tools are charts depicting the spot yield curve, the forward yield curve, and the breakeven levels. Each expiry (1M, 3M, 6M, 1Y) and strategy (payers, receivers, straddles) requires one of those charts. Figure 19.6, Figure 19.7, and Figure 19.8 show the example for 1Y expiry ATMF straddles, payer, and receiver swaptions on the JPY yield curve.



**FIGURE 19.6** Breakeven curves for 1Y ATMF straddles on different maturities of the JPY yield curve.

Sources: data – Bloomberg; chart – Authors.

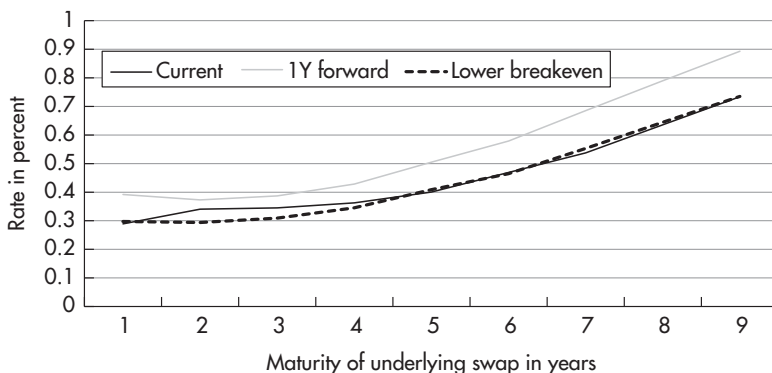
Data period: “Current” market data as of 18 Jun 2012.



**FIGURE 19.7** Breakeven curves for 1Y ATMF payer swaptions on different maturities of the JPY yield curve.

Sources: data – Bloomberg; chart – Authors.

Data period: “Current” market data as of 18 Jun 2012.



**FIGURE 19.8** Breakeven curves for 1Y ATMF receiver swaptions on different maturities of the JPY yield curve.

Sources: data – Bloomberg; chart – Authors.

Data period: “Current” market data as of 18 Jun 2012.

In this example, an analyst could answer the questions from above as follows:

- Looking at the straddle chart (Figure 19.6), the width of the breakeven band around the forward curve appears to be rather narrow. This corresponds to a relatively low implied volatility, thus a low option premium,

which translates into a narrow breakeven band. The conclusion is that expressing a macro view on yield levels through long option positions is attractive in general.

- In the next step, one can assess the different strategies (straddles, payers, receivers) versus each other. In the current example, one may conclude that given the low likelihood of BoJ rate hike expectations changing over the next year, the forward yields may well converge toward the current spot yield levels. This would imply a bias for lower rates and thus for long receiver swaption trades. This conclusion is confirmed by the breakeven chart for receivers (Figure 19.8), which shows that for 1Y rates and at the long end of the curve the current yield level is below the breakeven level of a long ATMF receiver. Hence, if the yield curve remained unchanged over the next year, buying a 1Y1Y or 1Y7Y receiver swaption would return a small overall profit.
- Finally, since the distance between the breakeven line and the current yield level in Figure 19.8 is maximal for 1Y and 7Y underlying swap maturities, buying a 1Y1Y or 1Y7Y receiver swaption seems best from a breakeven point of view. As the yield outlook is usually more certain for shorter maturities, the best risk/reward balance could be found in 1Y1Y receivers.

Additionally, one may want to assess whether the current distance between breakeven levels and spot rates is a unique opportunity or rather close to the historical average. This could be done by plotting a time series of that difference. In the current example, it may reveal that the potential profit from being long 1Y1Y receivers is relatively small both on an absolute basis (Figure 19.8) and from a historical point of view. Thus, investors might be enticed to enter those trades at the current levels only if they have a strong view that 1Y rates will decline further.

These simple breakeven charts already contain much of the information relevant for trading decisions: the roll-down (difference between forward and spot), the absolute level of implied volatility (width of breakeven bands around forward curve), and its relative distribution across the maturity spectrum. In the example above, this information could be combined to find (relatively small) value in long 1Y1Y receiver swaptions, offering a good roll-down even slightly below the breakeven band, which reflects low implied volatility in general and at the short end of the curve in particular.

On the other hand, we observed two shortcomings of this basic approach:

- It requires many breakeven charts, one for every expiry (e.g. 1M, 3M, 6M, 1Y) and for every strategy (at a minimum, straddles, receivers, and payers). This makes at least 12 charts, each of which features a yield

curve of maybe 10 points. More than one hundred potential trades present a challenge to the oversight of the analyst and become hard to assess informally. In the example above, how does the greater distance between current yield and lower breakeven (Figure 19.8) in 7Y relative to 1Y compare with the greater sensitivity of 7Y relative to 1Y to potential rate hike expectations? It would therefore be desirable to have a uniform and quantitative criterion to assess all potential trading strategies.

- Every option trade of type ① with a single underlying is a macro position, as it takes a view on the level of the underlying at expiry. Thus, the approach of the breakeven analysis to compare the option premium (via translation into breakeven levels) with the level of the underlying at expiry in essence links the option market through breakeven levels to the macroeconomic expectations of the analyst. In the charts, the comparison of the option market as expressed in breakeven levels to macroeconomic scenarios is done *outside* of the analysis tool (breakeven charts) and thus informally. For example, we argued above verbally: “Given the BoJ is very unlikely to hike soon, it is also unlikely for 1Y rates in a year’s time to exceed 0.3% [lower breakeven in Figure 19.8] and thus for a long 1Y1Y receiver swaption to lose money [from ending less ITM than its up-front premium requires to break even].”

The idea is therefore to solve both shortcomings together by linking the breakeven analysis to a uniform and quantifiable macroeconomic variable. This incorporates the informal macroeconomic assessment of breakeven levels into the framework of the analysis and allows us to compare different option strategies by their performance with regard to the relevant macroeconomic variable. The macroeconomic variable chosen depends on the goal of the analysis and on the views of the analyst. If he has a view on BoJ policy, he will choose the policy rate; if he has a view on CPI, he will choose CPI. For example, if a 1Y yield of 0.3% corresponds to a CPI level of  $-0.1$ , then he can argue that for the lower breakeven not to be reached, CPI in Japan would need to rise to above  $-0.1$  over the next year.<sup>12</sup>

A formal analysis linking the option market through breakeven levels to macroeconomic variables and scenarios can be done in two directions:

- Either the breakeven levels are translated into corresponding macroeconomic variables. In the example above, we have calculated the “option

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<sup>12</sup>This is for illustration purposes only and does not intend to make statements about the link of interest rates to CPI, etc. Actually, the recent correlation between JPY swap rates and macroeconomic variables such as CPI or seasonally adjusted GDP growth has been rather poor.

implied” (i.e. breakeven level implied) CPI figure. This indicates the macroeconomic scenario to which the current pricing in the option market corresponds. Now that the analyst can assess the pricing in the option market in macroeconomic terms, he can express his macro views through options, if he sees that the pricing in the option market does not match his economic expectations.

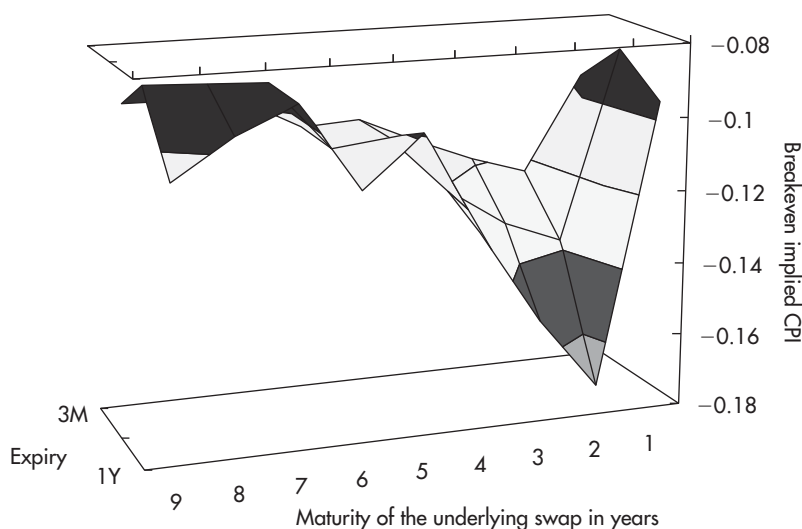
- Or, the other way round (which is more practicable when the macro model of the yield curve involves more than a single variable), one could define macroeconomic scenarios (baseline, recession, recovery), translate them via a macro model (like a VAR model) into yield forecasts, and compare these forecasts with the breakeven levels. If one finds an option strategy that benefits under each scenario, one has found a good candidate for a trade idea. This comparison of yield forecasts in different economic scenarios with breakeven levels could be done, for example, by calculating the number of standard deviations by which the breakeven level exceeds the yield forecast. Displaying these *t*-stats for different options simultaneously (e.g. in the form of a heat map for each macroeconomic scenario) can facilitate the selection of the best strategy.<sup>13</sup>

In the following, we provide an example of the first approach, assuming that we have a view on the Japanese CPI and want to compare the pricing of the swaption market with our CPI forecast. Note that this is for illustrative purposes only, as the correlation between CPI and interest rates in Japan (0.5) would, in our view, be too weak to base trades on.

In the first step, the breakeven levels for long receiver swaptions are calculated for each expiry (3M, 6M, etc.) and for each time to maturity of the underlying swap (1Y, 2Y, etc.). Then, the difference between that breakeven level and the current yield level is expressed in terms of the corresponding change in factor 1 of a PCA on the JPY swap yield curve.<sup>14</sup> As described in Chapter 3, the change in factor 1 can then be associated with a change in the CPI. As a result, for each expiry and underlying maturity of the receiver swaption, we obtain the CPI at which a long receiver position is expected to break even. In other words, we have translated the prices in the option market into the corresponding CPI figures. And since we have several expiries, we have even obtained the *evolution* of the CPI as implied by breakeven levels of receivers. Every underlying swap maturity results in a different implied CPI evolution, all of which are depicted in Figure 19.9.

<sup>13</sup>An example of this second approach can be found in “Front-End Payer Swaptions: Shorts Offer Value,” published by ABN AMRO Research on 25 February 2005.

<sup>14</sup>By dividing the yield difference in a certain maturity with the entry in the first eigenvector for that maturity.



**FIGURE 19.9** Option market-implied evolution of the CPI as calculated through the breakeven levels of long receiver swaptions.

Sources: data – Bloomberg; chart – Authors.

Data period: “Current” market data as of 18 Jun 2012.

Thus we find that a view on a low CPI can best be expressed through long 6M1Y receiver swaptions. Note the way the formal comparison of all option strategies through the uniform macroeconomic variable “breakeven implied CPI” has confirmed the informal argument we made verbally when looking at the breakeven charts. If in six months’ time the CPI is below  $-0.08$ , then the 1Y swap rate level is expected to be below the breakeven level of 30.4 bp (basis points), and hence the profit from a long 6M1Y receiver swaption at expiry is expected to be greater than the up-front premium payment required.

Likewise, a view on a high CPI can best be expressed through short 1Y2Y receiver swaptions. As long as the CPI in one year is above  $-0.17$ , the option premium received up-front is expected to exceed the potential loss from the option being exercised at expiry.

Furthermore, in case a tradable instrument is selected as the universal comparison for all breakeven levels (e.g. a commodity future rather than the CPI), this approach leads us to compare the volatility in the swaption market with the volatility in the option market on the tradable instrument (e.g. options on the commodity future). Thereby a relative value link between two option markets (and their implied macroeconomic scenarios) is established. This could be the basis for exploiting mismatches between the two



(e.g. via long swaptions versus short options on the commodity future). These relative value trades are hedged against the macroeconomic scenario actually materializing, just exploiting the different anticipation of macroeconomic scenarios in the two option markets.

## OPTION TRADE TYPE ①: TWO OR MORE UNDERLYINGS

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Option trade type ① on one single underlying gives a conditional expression of macro views (yield level), while option trade type ① on two or more underlyings gives a conditional expression of relative value views (e.g. yield curve steepness).

For example, we could express a 2Y-10Y curve flattening view via any of the three following alternatives:

- pay 2Y, receive 10Y (no options);
- sell a 2Y receiver, buy a 10Y receiver (swaption);
- buy a 2Y payer, sell a 10Y payer (swaption).

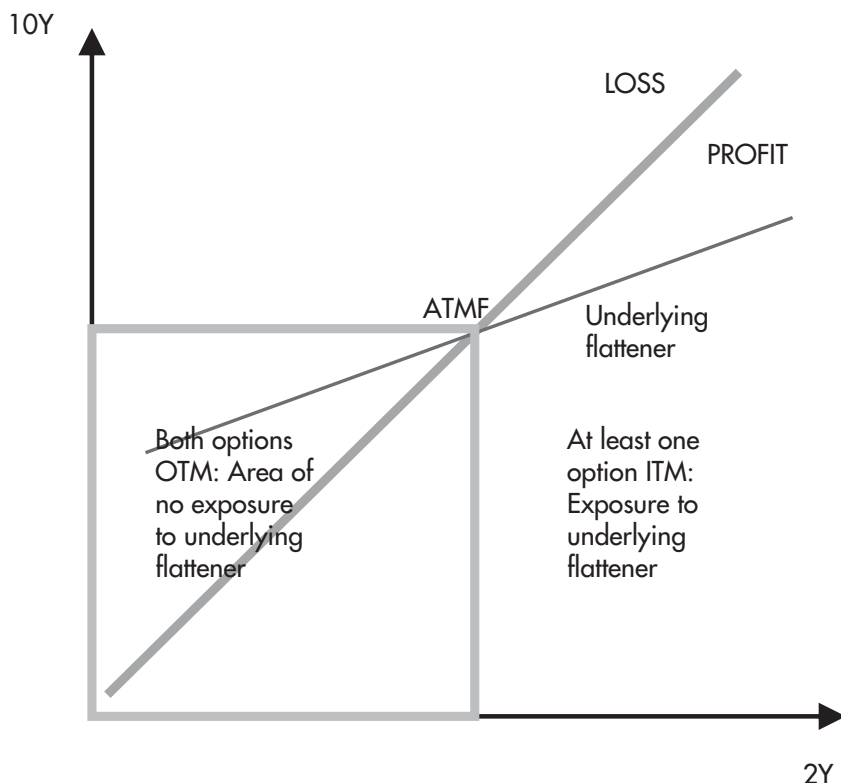
In comparison with the first alternative, the expression with options has the following specific features:

- (1) It is conditional (i.e. it is exposed to the underlying curve steepness only under the condition that the options are ITM at expiry).
- (2) It involves an option premium, which is determined by the difference in 2Y and 10Y implied volatility at entry.

And the two possibilities (via receivers or payers) to express the curve flattening through options are different in the way they offer (1) and (2):

- (1) The expression with receivers gives a curve position conditional on a rally, while the expression with payers gives a curve position conditional on a selloff.
- (2) The expression with receivers will return an up-front premium pick-up if implied 2Y volatility is higher than implied 10Y volatility (adjusted for hedge ratios); the expression with payers will return an upfront premium pick-up if implied 2Y volatility is lower than implied 10Y volatility.

The conditionality (1) allows selecting the directional environment that is favorable for the trade. For the following example, assume that the 2Y-10Y yield curve usually flattens in a selloff and steepens in a rally. Thus, we would prefer to express a curve-flattening view with *payer* swaptions: only in



**FIGURE 19.10** P&L from a 2Y-10Y curve-flattening position with swaps and with payer swaptions.

Source: Authors.

the case of a selloff would these options expire ITM, hence giving us exposure to a curve-flattening position, which is expected to perform well in a selloff. In the case of a rally (associated with curve steepening), both options expire OTM, thus preventing us from exposure to (underperforming) yield curve flattening. Illustrating this crucial point, Figure 19.10 depicts the way that conditionality of option trades of type ① can be used to select the favorable directional environment and to eliminate exposure to curve positions in unfavorable directional situations.

Note that for this purpose we want curve trades to be directional. Thus, we use here a basis point value (BPV) neutral weighting rather than a PCA-neutral weighting (see discussion of Figure 3.14). Put otherwise, when expressing curve trades with swaps (no options), we wanted to *solve* the